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Management of Patients With Diabetes and Chronic Kidney Disease

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Type 2 diabetes mellitus (T2D), and to a lesser extent type 1 (T1D), are rapidly increasing in incidence and remain the leading cause of chronic kidney disease (CKD) and end-stage kidney disease (ESKD) in the Western world.[1] This chapter reviews the diagnostic and management considerations of the patient with diabetes and CKD, referred to as diabetic kidney disease (DKD), focusing on strategies with the potential to improve clinical outcomes.

DIAGNOSTIC CONSIDERATIONS

Diabetic kidney disease is usually a clinical diagnosis in patients with diabetes and CKD, with kidney biopsy reserved for atypical presentations such as frank hematuria amid negative urologic investigations, nephrotic range albuminuria in the presence of normal glomerular filtration rate (GFR), or an accelerated progression of albuminuria or decline in GFR.[2] In such populations, one-half to two-thirds of patients have nondiabetic kidney disease.[3,4] The presence of diabetic retinopathy increases the pretest probability of DKD, but its absence is less helpful in ruling it out.[4,5]

GENERAL MANAGEMENT CONSIDERATIONS

Patients with DKD are at risk of premature mortality from cardiovascular (CV) disease.[6,7] The Kidney Disease: Improving Global Outcomes (KDIGO) guideline on Diabetes Management in CKD recommends lifestyle changes, identification and treatment of kidney-heart risk factors, patient self-management, and team-based integrated care programs to complement targeted pharmacotherapy (Fig. 33.1).[8] Key lifestyle changes include smoking cessation, salt restriction to less than 5 g/day (sodium <2 g/day); adoption of diets high in fruits, vegetables, unsaturated fats, and plant-based proteins while avoiding processed meats and carbohydrates; and moderate-intensity physical activity for at least 150 minutes weekly. Self-management education programs empower patients to actively participate in this comprehensive care model, aiming to slow DKD progression, reduce CV and treatment-related complications, and improve satisfaction with and adherence to treatment plans.[9] Future aims include halting the disease process and reversing damage.

A multidisciplinary approach is pivotal. It should include nephrologists, primary care providers, endocrinologists/diabetologists, cardiologists, nephrology/diabetes nurse specialists, podiatrists, ophthalmologists, and dieticians, as needed. Regular clinical monitoring improves outcomes in CKD patients,[10] with current recommendations for follow-up ranging from yearly to quarterly depending on the level of GFR level and its rate of decline.

MONITORING DIABETIC KIDNEY DISEASE

Beyond methods for measuring glycemic control (see later), albuminuria (typically quantified by albumin-to-creatinine ratio [ACR]) and estimated GFR (eGFR) constitute the mainstay of monitoring in DKD.

Measures of Glycemic Control

Hemoglobin A1c

Hemoglobin A1c (HbA1c) arises from nonenzymatic glycation of blood hemoglobin[11] and reflects average glycemia over approximately 10 to 12 weeks. Advantages are its accessibility and low intrasubject variability.[12] Limitations include inability to provide measures of short-term serum glucose levels or glycemic variability and reduced correlation with blood glucose levels in advanced CKD stages, due largely to increased red blood cell turnover (see Chapter 86).[13] Despite this, elevated HbA1c (>8% [>64 mmol/mol]) has been associated with higher all-cause and CV mortality in people with advanced DKD, including those treated with dialysis or kidney transplantation.[14-18]

Guidelines of the KDIGO recommend an individualized HbA1c target ranging between less than 6.5% and 8.0% or less in people with nondialysis DKD. Fig. 33.2 shows factors guiding decisions on individual HbA1c targets in DKD. In dialysis patients, the optimal glycemic target is unknown.

Self-Monitoring of Blood Glucose

In contrast to HbA1c, direct glucose measurements are not affected by severe CKD. Self-monitoring of blood glucose (SMBG) using a fingerstick and glucometer may facilitate maintenance of glycemic control. Self-monitoring is especially valuable in patients receiving insulin or other agents that may cause hypoglycemia. Because SMBG is at the discretion of the patient, episodes of hypoglycemia or hyperglycemia may be missed. Additionally, glucose variability cannot be readily assessed.

Continuous Glucose Monitoring

Continuous glucose monitoring (CGM) devices measure interstitial glucose using subcutaneous sensors. Depending on the system used, glucose measurements can be programmed to be performed continuously or at specific intervals (typically every 5–15 minutes). Some models allow for visualization of glucose levels or trends in real time. Devices can generate reports that display glucose trends over time (specifying time within, above, and below the target glucose range) in addition to measures of glucose variability. A typical target range is a glucose level of 70 to 180 mg/dL (3.9–10.0 mmol/L) in more than 70% of readings.[19]

Continuous glucose monitoring facilitates meeting glycemic targets and encourages self-management, with newer models offering linkage with insulin delivery systems or smartphone integration. This type

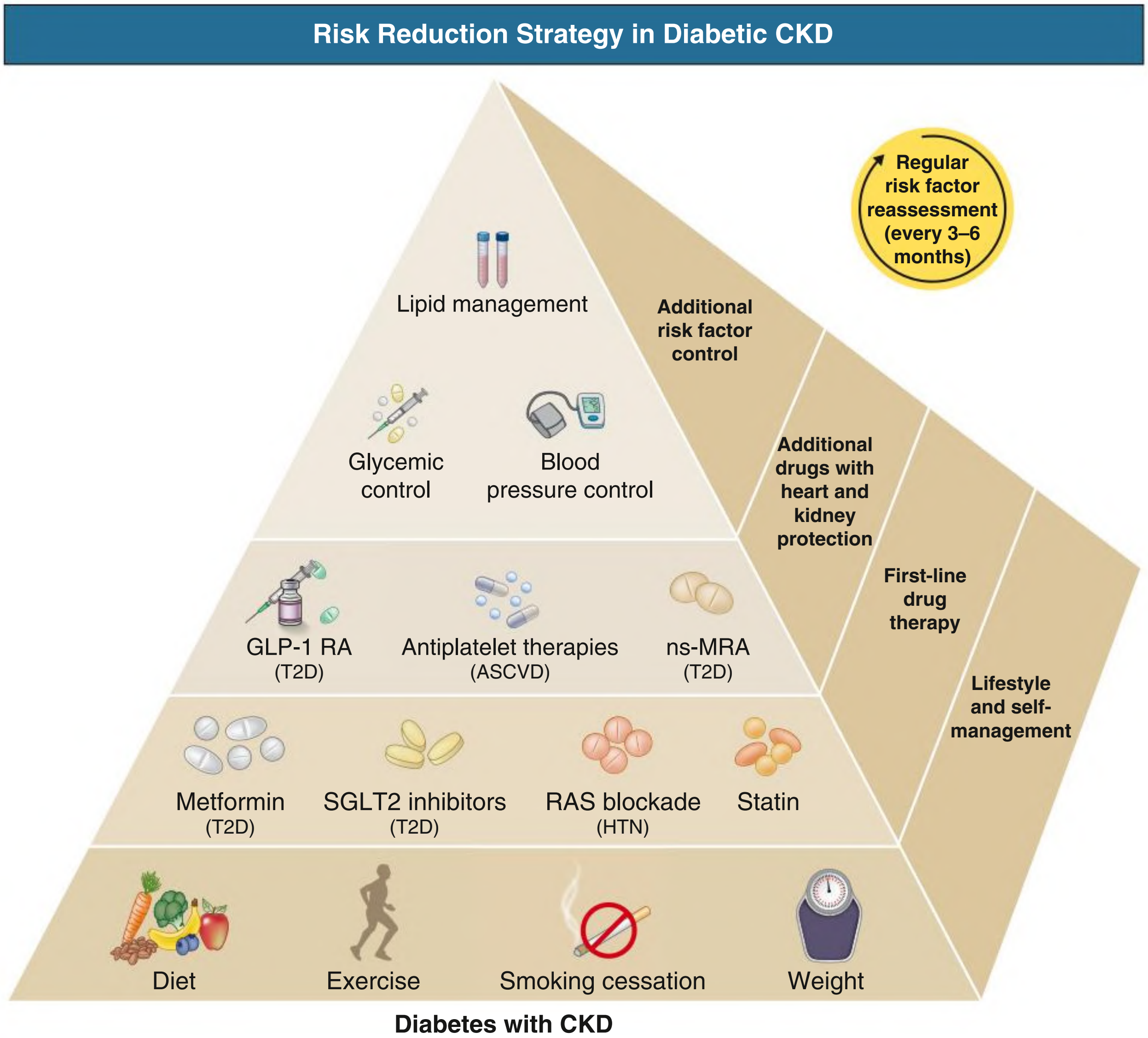

Fig. 33.1 Chronic Kidney Disease and Diabetes. Patients with diabetes and chronic kidney disease *(CKD)* Should be treated with a comprehensive strategy to reduce risks of kidney disease progression and adverse cardiovascular outcomes. Lifestyle and risk factor modification form the foundation of therapy, with pharmacotherapy targeted to appropriate patients using evidence from clinical trials. *RAS,* Renin-angiotensin system; *SGLT2,* sodium-glucose cotransporter-2. (From Kidney Disease: Improving Global Outcomes [KDIGO] Diabetes Work Group. KDIGO 2022 clinical practice guidelines for diabetes management in chronic kidney disease. *Kidney Int.* 2022;102:S1-S127.)

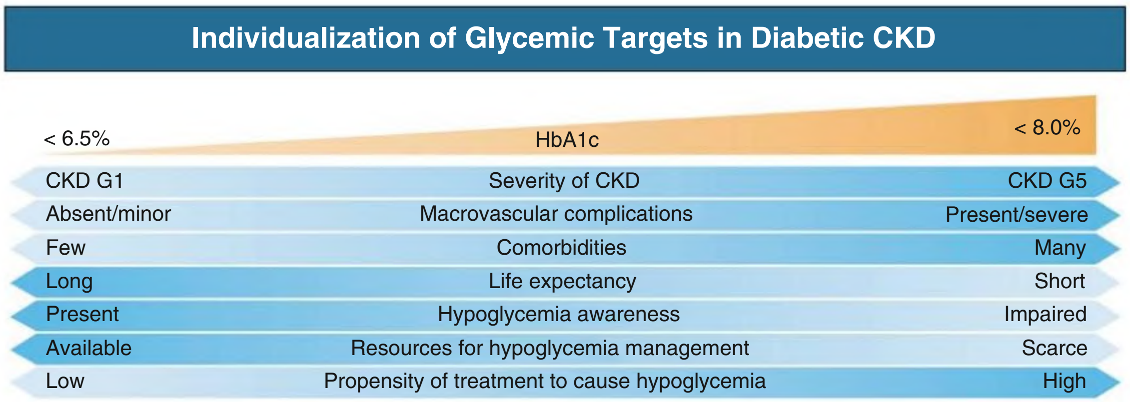

Fig. 33.2 Glycemic Targets Should be Individualized for Patients with Diabetes and Chronic Kidney Disease (CKD). Clinical factors affecting benefits and risks, depicted here, should be assessed to inform shared decision making. *CKD G1,* Estimated glomerular filtration rate (eGFR) ≤90 mL/min/1.73 m^2; *CKD G5,* eGFR <15 mL/min/1.73 m^2. *HbA1c,* Hemoglobin A1c. (From Kidney Disease: Improving Global Outcomes [KDIGO] Diabetes Work Group. 2022 clinical practice guidelines for diabetes management in chronic kidney disease. *Kidney Int.* 2022;102:S1-S127.)

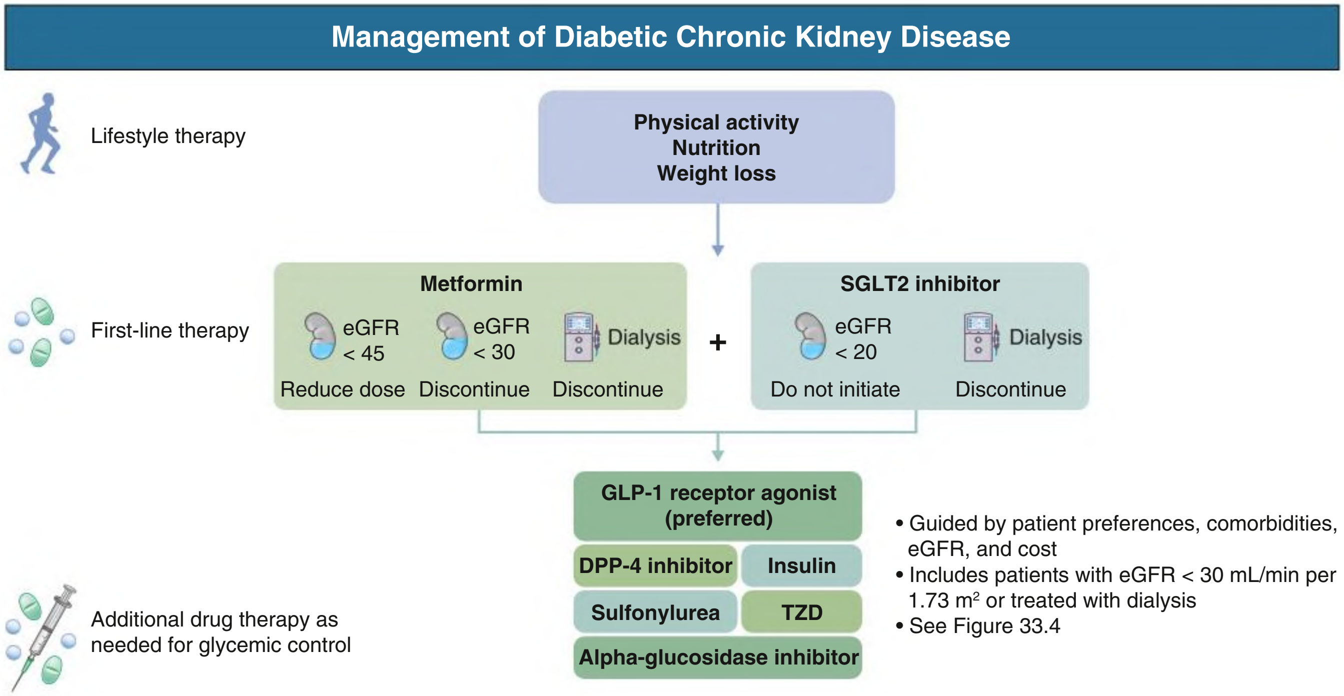

Fig. 33.3 Most patients with type 2 diabetes and chronic kidney disease should be treated with lifestyle therapy, metformin, and a sodium-glucose cotransporter-2 (SGLT2) inhibitor, depending on clinical factors and estimated glomerular filtration rate. Additional drug therapy may be needed to control glycemia if metformin and SGLT2 inhibition are not tolerated or contraindicated. *DPP-4,* Dipeptidyl peptidase-4; *eGFR,* estimated glomerular filtration rate; *GLP-1,* glucagon-like peptide 1; *TZD,* thiazolidinedione. (From Kidney Disease: Improving Global Outcomes [KDIGO] Diabetes Work Group. 2022 clinical practice guidelines for diabetes management in chronic kidney disease. *Kidney Int.* 2022;102:S1-S127.)

of monitoring is especially valuable for people who have highly variable glucose levels, are at risk for experiencing severe hypoglycemia, and those for whom HbA1c levels do not align with measured blood glucose. Disadvantages of CGM include skin irritation related to the device patch and potentially high cost.

Albuminuria

Albuminuria is classified as normoalbuminuria (<30 mg albumin/24 h), moderately increased albuminuria (30–300 mg/24 h), and severely increased albuminuria (>300 mg/24 h). The terms microalbuminuria and macroalbuminuria should be abandoned. Any albuminuria is associated with an increased risk of DKD progression and cardiovascular disease (CVD).[20,21] Moderately increased albuminuria was long viewed a first step in DKD, leading to worsening albuminuria, eGFR loss, and ESKD.[22] However, reduced GFR in the absence of albuminuria is common, particularly in T2D, with up to 57% of patients in certain CKD cohorts presenting in this manner.[23-25] These latter patients are still at increased risk of progression, albeit with slower rates of eGFR decline than albuminuric DKD.[26] Additionally, albuminuria in patients with moderately increased albuminuria may not progress to severely increased albuminuria and even undergo spontaneous remission.[27,28]

Glomerular Filtration Rate

Measured and estimated GFR are important predictors of kidney prognosis, particularly when combined with albuminuria. In healthy persons the annual age-related eGFR decline is 0.5 to 1 mL/min/1.73 m^2.[29] A large cohort of participants with T1D (mean duration of 5.5 years) had median decline in eGFR of 3 mL/min/1.73 m^2 over 3 years, with 25% experiencing a 3-year decline of 14 mL/min/1.73 m^2 or greater.[30] In more contemporary DKD cohorts, albuminuria is still associated with rapid eGFR decline even in the presence of renin-angiotensin system (RAS) inhibition. For example, the high kidney-risk CREDENCE trial participants with T2D and DKD experienced an annual eGFR decline of 4.71 ± 0.15 mL/min/1.73 m^2 per year.[31] Limitations of using GFR to monitor disease progression include nonlinear trajectories in GFR decline, including the occurrence of hyperfiltration early in the disease course and treatment-induced hemodynamic changes in kidney function, imperfect correlations between estimated and measured GFR, and potential underestimation of eGFR decline in clinical trial settings when compared with real-world clinical scenarios.

Models to Predict Kidney Failure Risk

The Kidney Failure Risk Equation (KFRE) is a widely used prediction instrument that has been validated in several cohorts and CKD etiologies, including DKD (see https://kidneyfailurerisk.com).[32] The KFRE uses patient age, sex, eGFR, and ACR to offer 2-year and 5-year risk estimates of developing ESKD. In addition to helping nephrologists quickly convey prognostic information to patients, KFRE-derived risk estimates are routinely used to anticipate and plan for the initiation of kidney-replacement therapies.

MANAGEMENT OF GLYCEMIA

Insulin is the primary treatment for T1D, with dosing schedules iteratively refined based on observed blood glucose concentrations at all levels of kidney disease. In T2D, oral antihyperglycemic agents are preferred as foundational therapy because of their lower risks of hypoglycemia and, for some drug classes, reduction in the risk of progressive CKD or CV events. For patients with T2D and CKD, lifestyle treatments, metformin, and sodium-glucose cotransporter-2 inhibitors (SGLT2i) are recommended as first-line therapy, with some restrictions according to eGFR (Fig. 33.3). Additional or alternate agents should be selected according to patient preferences, comorbidities, and other characteristics when needed to achieve glycemic goals (Fig. 33.4).

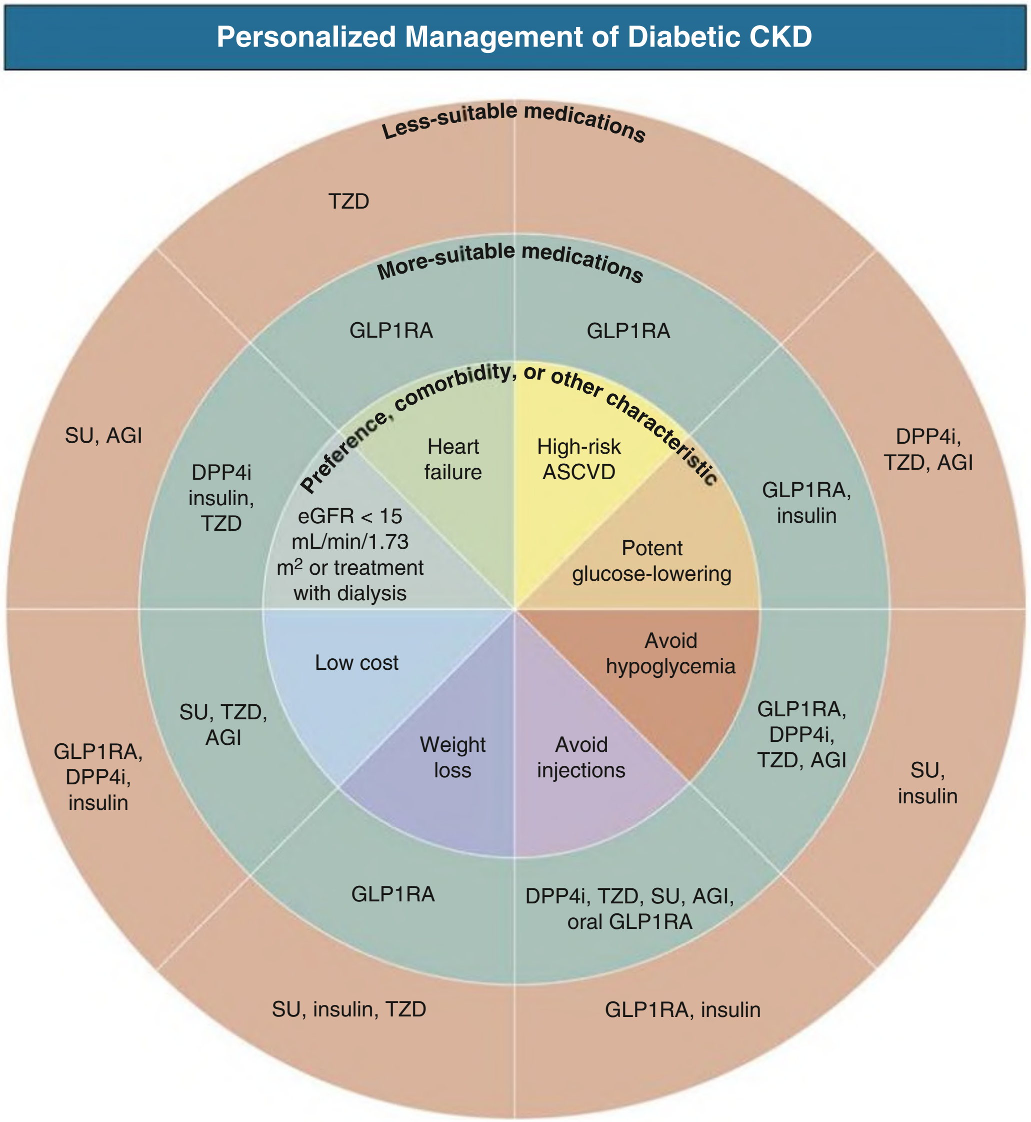

Fig. 33.4 For patients with type 2 diabetes and chronic kidney disease (CKD) who have not achieved individualized glycemic targets despite use of metformin and sodium-glucose cotransporter-2 inhibitor treatment, or who are unable to use those medications, patient preferences, comorbidities, and other characteristics should be used to select additional or alternate glucose-lowering medications. In general, glucagon-like peptide 1 receptor agonists (GLP1RA) may be preferred because of their beneficial cardiovascular effects, and possible beneficial effects on CKD progression. *AGI,* Alpha-glucosidase inhibitor; *ASCVD,* atherosclerotic cardiovascular disease; *DPP-4i,* dipeptidyl peptidase-4 inhibitor; *eGFR,* estimated glomerular filtration rate; *SU,* sulfonylurea; *TZD,* thiazolidinedione. (From Kidney Disease: Improving Global Outcomes [KDIGO] Work Group. 2022 clinical practice guidelines for diabetes management in chronic kidney disease. *Kidney Int.* 2022;102:S1-S127.)

Hypoglycemia and Hyperglycemia

Episodes of hypoglycemia are more common in patients with severe CKD or requiring dialysis due to reduced insulin clearance and impaired renal gluconeogenesis. Hypoglycemic episodes may result in increased short-term mortality, and these risks may be further elevated in patients who are elderly or have multiple comorbidities.[33] In stage III CKD, regular review of the oral antihyperglycemic regimen or insulin dosing is essential. With progression of CKD, medications that have a higher risk of inducing hypoglycemia should be dose reduced or replaced based on the following considerations.

Oral Hypoglycemic Agents

Many oral agents for the treatment of T2D require dose adjustments with reduced GFR (Table 33.1). Also, in advanced CKD, drug-drug interactions increase together with further complications of diabetes. For example, gastroparesis affects the pharmacokinetics of oral medications.

Biguanides

Metformin decreases hepatic glucose production, increases insulin sensitivity and insulin-mediated utilization of glucose in peripheral tissues, and decreases intestinal glucose absorption. Metformin is recommended as the first-line agent for management of T2D in combination with SGLT2i and lifestyle measures (Fig. 33.3).[34] In addition to effectively improving glycemic control, metformin modestly reduces the risk of CV events and death in T2D.[35] Metformin dosage should be reduced when eGFR drops below 45 mL/min/1.73 m^2, and such patients should be instructed to temporarily stop metformin in states

TABLE 33.1 Daily Dosing for Selected Oral Hypoglycemic Agents

Class	Drug	CKD1 and 2: eGFR >60 mL/min/1.73 m²	CKD3a: eGFR 45–59 mL/min/1.73 m²	CKD3b: eGFR 30–44 mL/min/1.73 m²	CKD4: eGFR 15–29 mL/min/1.73 m²	CKD5: eGFR <15 mL/min/1.73 m² or Dialysis
Biguanides	Metformin IR	No adjustment	No adjustment	850 mg or 1000 mg once daily	Avoid	Avoid
	Metformin ER	No adjustment	No adjustment	1000 mg once daily	Avoid	Avoid
SGLT-2i	Canagliflozin	No adjustment	100 mg once daily	100 mg once daily	Avoid initiation	Avoid
	Dapagliflozin	No adjustment	No adjustment	No adjustment	Avoid initiation with eGFR <25 mL/min/1.73 m²[a]	Avoid
	Empagliflozin	No adjustment	No adjustment	No adjustment	Avoid initiation with eGFR <20 mL/min/1.73 m²[a]	Avoid
Incretin mimetics (GLP-1 receptor agonists)	Exenatide IR	No adjustment	No adjustment	No adjustment	Avoid	Avoid
	Exenatide ER	No adjustment	No adjustment	Avoid	Avoid	Avoid
	Dulaglutide	No adjustment	No adjustment	No adjustment	No adjustment	Avoid
	Semaglutide (injection or oral)	No adjustment	No adjustment	No adjustment	Limited experience	Limited experience
	Liraglutide	No adjustment	No adjustment	No adjustment	Limited experience	Limited experience
Gliptins (DPP-4 inhibitors)	Linagliptin	No adjustment	No adjustment	No adjustment	No adjustment	No adjustment
	Sitagliptin	No adjustment	No adjustment	50 mg once daily	25 mg once daily	25 mg once daily
	Saxagliptin	No adjustment	No adjustment	2.5 mg once daily	2.5 mg once daily	2.5 mg once daily
Thiazolidinediones	Pioglitazone	No adjustment	No adjustment	No adjustment	No adjustment	No adjustment
Sulfonylureas	Gliclazide	40 mg once daily	40 mg once daily	40 mg once daily	40 mg once daily	Avoid
	Glipizide	No adjustment	20 mg once daily	20 mg once daily	20 mg once daily	Avoid
	Glimepiride	No adjustment	1 mg once daily	1 mg once daily	1 mg once daily	Avoid
Meglitinides	Repaglinide	No adjustment	No adjustment	No adjustment	No adjustment	Limited experience
α-Glucosidase inhibitors	Acarbose	No adjustment	No adjustment	No adjustment	Avoid	Avoid

[a]Threshold levels of eGFR for drug initiation are based on published clinical trials. Regulatory guidance may vary by agency, over time, and according to specific indication.

CKD, Chronic kidney disease; *DPP-4,* dipeptidyl peptidase-4; *eGFR,* estimated glomerular filtration rate; *ER,* extended release; *FDA,* Food and Drug Administration; *GLP-1,* glucagon-like peptide-1; *IR,* immediate release; *SGLT-2,* sodium-glucose cotransporter-2.

Listed dosages reflect the maximum daily recommendations based on manufacturer recommendations and expert opinion. Drug initiation and uptitration should be done slowly and cautiously with declining eGFR. Available formulations and pharmacologic guidelines vary by country.

of dehydration, before the administration of radiocontrast media, and other situations where there is an increased risk for acute kidney injury (AKI). Metformin is not recommended for people with eGFR less than 30 mL/min/1.73 m² because of the risk for lactic acidosis. However, a Cochrane review[36] found no evidence that advanced CKD was associated with increased risk for lactic acidosis in patients receiving appropriate doses of metformin.[35,37]

Sodium-Glucose Cotransporter-2 Inhibitors

SGLT2i block kidney proximal tubular glucose reabsorption. Resulting reductions in blood glucose are greater with higher concentrations of blood glucose and higher GFR. Along with metformin, SGLT2i are first-line antihyperglycemic therapy in DKD, given their cardioprotective and renoprotective effects (Fig. 33.3). In the EMPA-REG OUTCOME trial in patients with T2D and eGFR above 30 mL/min/1.73 m², empagliflozin reduced the risk of CV mortality, all-cause mortality, and hospitalization for congestive heart failure with favorable effects on weight and systolic blood pressure. The EMPA-REG OUTCOME trial also reported a reduced progression of kidney failure and albuminuria.[38] The CREDENCE trial enrolled patients with T2D, eGFR 30 to 90 mL/min/1.73 m², and macroalbuminuria and demonstrated reduced CKD progression and incident kidney failure with canagliflozin.[31] Additional randomized controlled trials (RCTs; CANVAS, DECLARE-TIMI 58, VERTIS, DAPA-CKD, DAPA-HF, EMPEROR-Reduced, SCORED, SOLOIST) have consistently demonstrated substantial long-term CV and kidney benefits of SGLT2i across eGFR despite reduced antihyperglycemic effects (Table 33.2).[39-46]

The mechanisms underlying kidney benefits of SGLT2i are incompletely understood. Effects have been attributed in part to restoration of tubuloglomerular feedback and reduction of glomerular hyperfiltration via increased distal sodium delivery to the macula densa. Other mechanisms may include improvements in energy metabolism, with a shift from carbohydrate to ketone metabolism inducing an energy-efficient mitochondrial oxygen consumption that reduces kidney hypoxia.[47] Reduced glomerular hyperfiltration is thought to account for the acute decline in eGFR observed with SGLT2i initiation.

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TABLE 33.2 Summary of Selected Large Clinical Trials of Recently Established and Potential New Therapeutics in Diabetic Kidney Disease

Study	Treatment Arms	Duration	Patient Cohort	Outcome
SGLT2i				
Evaluation of the Effects of Canagliflozin on Renal and Cardiovascular Outcomes in Participants with Diabetic Nephropathy (CREDENCE),[31] NCT02065791	Canagliflozin 100 mg daily vs. placebo	2.6 years	4401 T2D patients with stage 2 or 3 CKD and macroalbuminuria and on an ACE inhibitor or ARB	**Primary:** 30% reduction in ESKD, S-creatinine doubling, renal/CV death **Secondary:** 20% reduction in CV death, nonfatal MI, nonfatal stroke; 39% reduction in hospitalization for CHF; 34% reduction in composite renal endpoint (ESKD, doubling of serum Cr, renal death)
Empagliflozin, Cardiovascular Outcomes, and Mortality in Type 2 Diabetes (EMPA-REG OUTCOME),[38] NCT01131676	Empagliflozin vs. placebo	3.1 years	7020 participants with T2D at high CV risk	14% reduction in composite; death from CV causes, nonfatal MI, or nonfatal stroke
Canagliflozin and Cardiovascular and Renal Events in Type 2 Diabetes (CANVAS),[39] NCT01032629	Canagliflozin vs. placebo	3.6 years	10,142 participants with T2D and high CV risk	**Primary:** 14% reduction in composite death from CV causes, nonfatal MI, nonfatal stroke **Secondary:** 27% reduction in albuminuria progression; 40% reduction in composite sustained 40% reduction in eGFR, need for renal replacement therapy, death from renal causes
Dapagliflozin and Cardiovascular Outcomes in Type 2 Diabetes (DECLARE-TIMI 58),[40] NCT01730534	Dapagliflozin vs. placebo	4.2 years	17,160 participants with T2D at risk for atherosclerotic CVD	**Primary:** 17% reduction in CV death or hospitalization for HF; no difference with respect to MACEs **Secondary:** 24% reduction in composite renal events (>40% decrease in eGFR to <60 mL/min/1.73 m^2, new ESKD, or death from renal or CV causes); 7% reduction in death from any cause
Cardiovascular Outcomes with Ertugliflozin (VERTIS),[41] NCT01986881	Ertugliflozin vs. placebo	3.5 years	8246 participants with T2D and atherosclerotic CVD	**Primary:** noninferiority for MACEs **Secondary:** no difference in death from CV causes or hospitalization for HF; no difference in death from renal causes, renal replacement therapy, or doubling of serum Cr
Dapagliflozin in Patients with Heart Failure and Reduced Ejection Fraction (DAPA-HF),[43] NCT03036124	Dapagliflozin vs. placebo	1.5 years	4744 participants with HF and an ejection fraction ≤40%	26% reduction in composite worsening HF or CV death
Cardiovascular and Renal Outcomes with Empagliflozin in Heart Failure (EMPEROR-Reduced),[44] NCT03057977	Empagliflozin vs. placebo	1.3 years	3730 participants with HF	**Primary:** 25% reduction in composite CV death or hospitalization for worsening HF **Secondary:** slower eGFR decline, reduced renal outcomes
Sotagliflozin in Patients with Diabetes and Chronic Kidney Disease (SCORED),[45] NCT03315143	Sotagliflozin vs. placebo	1.3 years	10,584 participants with T2D and CKD at risk for CVD	26% reduction in composite death from CV causes, HF hospitalizations/urgent visits; 16% reduction in death from CV causes, nonfatal MI, or nonfatal stroke
Sotagliflozin in Patients with Diabetes and Recent Worsening Heart Failure (SOLOIST),[46] NCT03521934	Sotagliflozin vs. placebo	9 months	1222 participants with T2D and a recent HF hospitalization	33% reduction in death from CV causes, hospitalization, and urgent visits for HF
DAPA-CKD,[42] NCT02065791	Dapagliflozin vs. placebo	2.4 years	T2D with DKD or nondiabetic kidney disease with eGFR 25–75 mL/min/1.73 m^2 and UACR 200–5000 mg/g	**Primary:** 44% reduction in kidney composite endpoint (≥50% sustained decline in eGFR, ESKD, or kidney or CVD death) **Secondary:** 29% reduction in CV mortality or hospitalization for CHF; 31% reduction in all-cause mortality

TABLE 33.2 Summary of Selected Large Clinical Trials of Recently Established and Potential New Therapeutics in Diabetic Kidney Disease—cont'd

Study	Treatment Arms	Duration	Patient Cohort	Outcome
EMPA-KIDNEY (NCT03594110)	Empagliflozin vs. placebo	Ongoing	DKD (T2D or T1D) or nondiabetic kidney disease with eGFR 20–45 mL/min/1.73 m^2 *or* eGFR 45–90 mL/min/1.73 m^2 with UACR ≥200 mg/g	**Ongoing:** Composite outcome of time to first occurrence of kidney disease progression (ESKD, sustained decline in eGFR to <10 mL/min/1.73 m^2), kidney death, or a sustained decline of ≥40% in eGFR from randomization), or CV death
GLP-1 Receptor Agonists				
Liraglutide and Cardiovascular Outcomes in Type 2 Diabetes (LEADER),[48] NCT01179048	Liraglutide vs. placebo	3.8 years	9340 participants with T2D at high CV risk	Reduction in composite of death from CV causes, nonfatal MI, or nonfatal stroke
Semaglutide and Cardiovascular Outcomes in Patients with Type 2 Diabetes (SUSTAIN-6),[49] NCT01720446	Semaglutide vs. placebo	2 years	3297 participants with T2D	**Primary:** 26% reduction in composite of CV death, nonfatal MI, or nonfatal stroke **Secondary:** reduction in new or worsening kidney dysfunction
Albiglutide and Cardiovascular Outcomes in Patients with Type 2 Diabetes and Cardiovascular Disease (HARMONY),[50] NCT02465515	Albiglutide vs. placebo	1.5 years	9463 participants with T2D and CVD	22% reduction in composite of CV death, MI, or stroke
Dulaglutide and Cardiovascular Outcomes in Type 2 Diabetes (REWIND),[51] NCT01394952	Dulaglutide vs. placebo	5.4 years	9901 participants with T2D and CV risk factors	12% reduction in composite nonfatal MI, nonfatal stroke, or death from CV causes
Effect of Semaglutide Versus Placebo on the Progression of Renal Impairment in Subjects With Type 2 Diabetes and Chronic Kidney Disease (FLOW), NCT03819153	Semaglutide vs. placebo	Ongoing	T2D with eGFR ≥50 to <75 mL/min/1.73 m^2 with UACR >300 mg/g and <5000 mg/g *or* eGFR ≥25 to <50 mL/min/1.73 m^2 with UACR >100 mg/g to <5000 mg/g	**Ongoing:** Time to first occurrence of a composite primary outcome event defined as persistent eGFR decline of ≥50% from trial start, reaching ESKD, death from kidney disease, or death from CVD
A Phase IIb, Multicenter, Randomised, Double-Blind, Placebo-Controlled, and Open-Label Comparator Study of Cotadutide in Participants Who Have Chronic Kidney Disease With Type 2 Diabetes Mellitus, (NCT04515849)	Cotadutide (dual GLP1 and glucagon receptor agonist) vs. placebo	Ongoing	T2D with eGFR ≥20 to <90 mL/min/1.73 m^2 with UACR >50 mg/g	**Ongoing:** UACR, HbA1c, body weight
PKCβ Inhibitors				
Treatment of Peripheral Neuropathy in Patients with Diabetes,[109] NCT00044421	Ruboxistaurin mesylate 32 mg vs. placebo	2.7 years	707 T2D participants with diabetic neuropathy	Patients treated with ruboxistaurin had lower UACR and higher eGFR
Antiinflammatory Agents				
Canakinumab Anti-Inflammatory Thrombosis Outcome Study (CANTOS),[110] NCT01327846	Canakinumab 300 mg vs. canakinumab 150 mg vs. placebo	3.7 years	10,061 adults with a history of MI and systemic inflammation (elevated high-sensitivity CRP >2 mg/mL); 40% had T2D and 46% of those with CKD in the trial had T2D	15% reduction in 3-point MACE: nonfatal MI, any nonfatal stroke, or CV death; subsequent post hoc analyses demonstrated that the risk of MACE was reduced in people with CKD and in those with albuminuria or diabetes

Continued

TABLE 33.2 Summary of Selected Large Clinical Trials of Recently Established and Potential New Therapeutics in Diabetic Kidney Disease—cont'd

Study	Treatment Arms	Duration	Patient Cohort	Outcome
A Study to Evaluate the Safety and Efficacy of CCX140-B in Subjects With Diabetic Nephropathy,[111] NCT01447147	CCX140-B 10 mg (CCR2 inhibitor) vs CCX140-B 5 mg vs. placebo	52 weeks	332 T2D patients with proteinuria, GFR ≥25 mL/min/1.73 m^2	Albuminuria lowering
Effects of Selonsertib in Patients with Diabetic Kidney Disease,[112] NCT02177786	1:1:1:1 allocation to selonsertib (oral daily doses of 2, 6, or 18 mg) or placebo	48 weeks	333 adults with moderate to advanced DKD (eGFR of 15–60 mL/min/1.73 m^2 at screening) and albuminuria, defined as a UACR ≥600 mg/g if stage 3a CKD, UACR ≥300 mg/g if stage 3b CKD, and UACR ≥150 mg/g if stage 4 CKD	**Primary endpoint:** no difference in eGFR at 48 weeks In post hoc analyses from 4 and 48 weeks, eGFR decline was reduced by 71% for the 18-mg group vs. placebo (difference 3.11 mL/min/1.73 m^2 per year, $P = .043$) Effects on UACR did not differ between selonsertib and placebo
Mineralocorticoid Receptor Antagonists				
Mineralocorticoid Receptor Antagonist Tolerability Study–Heart Failure (ARTS-HF),[113] NCT01807221	Finerenone (multiple doses) vs. eplerenone	90 days	1066 patients with worsening HFrEF and CKD and/or T2D	Finerenone reduced a composite endpoint of death from any cause, CV hospitalizations, or emergency presentation for worsening HF; reduced albuminuria
ARTS–Diabetic Nephropathy (RTS-DN),[114] NCT1874431	Finerenone (multiple doses) vs. placebo	90 days	823 T2D patients with high or very high albuminuria on ACEs or ARBs	Reduced albuminuria
Efficacy and Safety of Finerenone in Subjects with Type 2 Diabetes Mellitus and Diabetic Kidney Disease (FIDELIO-DKD),[83] NCT02540993	Finerenone vs. placebo	2.6 years	5734 T2D patients with DKD (persistent high albuminuria or very high albuminuria)	**Primary:** 18% reduction in time to first occurrence of the composite of onset of kidney failure, sustained decrease of eGFR ≥40% from baseline over at least 4 weeks, and renal death **Secondary:** 14% reduction in death from CV causes, nonfatal MI, nonfatal stroke, or hospitalization for CHF
Efficacy and Safety of Finerenone in Subjects with Type 2 Diabetes Mellitus and a Clinical Diagnosis of Diabetic Kidney Disease (FIGARO-DKD), NCT02540993	Finerenone (10 mg vs. 20 mg vs. placebo)	3.4 years	7437 T2D patients with DKD (persistent high albuminuria or very high albuminuria)	**Primary:** 13% reduction in time to first occurence of the composite of death from CV causes, nonfatal MI, nonfatal stroke, or hospitalization for CHD.Secondary outcome: no significant difference in time to kidney failure, sustained decrease of eGFR ≥40% from baseline, or renal death.
Endothelin Receptor Antagonists				
SONAR,[115] NCT01858532	Atrasentan 0.75 mg vs. placebo	2.2 years	2,648 patients with T2D, eGFR 25–75 mL/min/1.73 m^2 and ACR 300–500 mg/g on RAS blockers	Early termination (lower number of events), 35% relative risk reduction of doubling of serum Cr or ESKD
Neprilysin Inhibitors				
Prospective Comparison of ARNI with ACEI to Determine Impact on Global Mortality and Morbidity in Heart Failure trial,[116] (PARADIGM-HF)	Sacubitril/valsartan vs. enalapril	2.25 years	8399 patients with HFrEF with and without DM	Slower rate of eGFR decline with neprilysin inhibition (–1.3 vs. –1.8 mL/min/1.73 m^2 per year; $P < .0001$); though greater increase in UACR (1.20 mg/mmol [95% CI, 1.04–1.36 mg/mmol] vs. 0.90 mg/mmol [95% CI, 0.77–1.03 mg/mmol]); greater impact in diabetes cohort

TABLE 33.2 Summary of Selected Large Clinical Trials of Recently Established and Potential New Therapeutics in Diabetic Kidney Disease—cont'd

Study	Treatment Arms	Duration	Patient Cohort	Outcome
Efficacy and Safety of LCZ696 Compared with Valsartan, on Morbidity and Mortality in Heart Failure Patients With Preserved Ejection Fraction trial,[117] (PARAGON-HF)	Sacubitril/valsartan vs. valsartan	2.9 years	4822 with HFpEF with and without DM	Nonsignificant reduction in the primary outcome of composite total hospitalizations for CHF and death from CV causes (HR, 0.87; 95% CI, 0.75–1.01; $P = .06$); renal composite outcome (time to first occurrence of either ≥50% reduction in eGFR, ESKD, or death from renal causes [HR, 0.50; 95% CI, 0.33–0.77; $P = .001$])
United Kingdom Heart and Renal Protection-III[118]	Sacubitril/valsartan vs. irbesartan	12 months	414 participants (40% with a history of DM at baseline) with eGFR 20–60 mL/min/1.73 m^2 and without CHF	Sacubitril/valsartan reduced blood pressure, NT-pro-BNP, and troponin I significantly more compared with irbesartan, but without affecting eGFR or UACR

ACE, Angiotensin-converting enzyme inhibitor; *ACR,* albumin-to-creatinine ratio; *ARB,* angiotensin receptor blocker; *CHF,* congestive heart failure; *CI,* confidence interval; *CKD,* chronic kidney disease; *Cr,* creatinine; *CRP,* C-reactive protein; *CV,* cardiovascular; *CVD,* cardiovascular disease; *DKD,* diabetic kidney disease; *DM,* diabetes mellitus; *eGFR,* estimated glomerular filtration rate; *ESKD,* end-stage kidney disease; *GLP-1,* glucagon-like peptide-1; *HbA1c,* hemoglobin A1c; *HF,* heart failure; *HFpEF,* heart failure with preserved ejection fraction; *HFrEF,* heart failure with reduced ejection fraction; *HR,* hazard ratio; *MACE,* major adverse cardiovascular event; *MI,* myocardial infarction; *PKCβ,* protein kinase Cβ; *RAS,* renin-angiotensin system; *SGLT-2i,* sodium-glucose cotransporter-2 inhibitor; *T1D,* type 1 diabetes; *T2D,* type 2 diabetes; *UACR,* urinary albumin-to-creatinine ratio.

SGLT2i are associated with increased risk of euglycemic diabetic ketoacidosis and genital mycotic infections. Patients should be advised of these adverse effects and monitored closely for their development. Additionally, the SGLT2i class is part of the routine "sick day" medication list that should be held during periods of illness.

Incretin Mimetics: Glucagon-Like Peptide-1 Receptor Agonists

Glucagon-like peptide-1 receptor agonists (GLP-1 RAs) promote glucose-mediated insulin secretion by pancreatic β-cells in response to food entering the gut and suppress glucagon secretion. These receptor agonists are potent antihyperglycemic agents and stimulate weight loss by appetite suppression, both centrally and by affecting gastric motility. Guidelines of KDIGO recommend initiation of GLP-1 RA in T2D patients who have not achieved glycemic control despite metformin and SGLT2i, in large part because of these agents' CV and kidney benefits. In RCTs, GLP-1 RA therapies reduced major adverse CV events in people with T2D at high risk for CVD (LEADER, SUSTAIN-6, HARMONY, REWIND; Table 33.2).[48-51] A subset of these studies reported a reduction in albuminuria and eGFR decline with GLP-1 RA use (detailed further later). Adverse effects associated with GLP-1 RA use include gastrointestinal symptoms and, with subcutaneously administered formulations, injection site reactions. Glucagon-like peptide-1 receptor agonists should be avoided in people with a history of medullary thyroid cancer or pancreatitis.

Sulfonylureas

Sulfonylureas are a class of insulin secretagogues that stimulate pancreatic insulin secretion by closing potassium–adenosine triphosphate channels on β-cell plasma membranes. First-generation sulfonylureas are long acting and should be avoided in people with CKD given their almost exclusive renal excretion. Second-generation agents are short acting and primarily metabolized by the liver, with most metabolites (which hold antihyperglycemic properties) undergoing kidney clearance. Their use carries the risk for hypoglycemia, especially as eGFR declines. Sulfonylureas are highly protein bound but can be displaced into the circulation by other drugs such as salicylates or β-blockers, further contributing to hypoglycemia. Sulfonylureas are therefore discouraged in patients with severe CKD.

Thiazolidinediones

Thiazolidinediones are peroxisome proliferator-activated receptor (PPAR) modulators that function by increasing insulin sensitivity. These agents do not cause hypoglycemia and come at a low cost. However, thiazolidinediones may cause weight gain and fluid retention through transcriptional upregulation of tubular amiloride-sensitive sodium channels, which may be problematic in people with CKD at risk for heart failure. Higher rates of bone fractures also have been reported in patients treated with these agents.

Meglitinides

Meglitinides are primarily metabolized in the liver and act as insulin secretagogues similar to sulfonylureas. Because of the risk of hypoglycemia, these agents should be avoided with advanced CKD. Alternatively, formulations that are not primarily excreted by the kidneys, such as repaglinide, may be used.

Gliptins: Dipeptidyl Peptidase-4 Inhibitors

Gliptins inhibit the effect of dipeptidyl peptidase-4 (DPP-4), a cellular membrane protein expressed in a variety of tissues that rapidly degrades endogenous incretin hormones (e.g., GLP-1). Agents such as linagliptin are primarily metabolized by the liver and excreted in the bile, and thus they do not require dose adjustments in CKD. Gliptins are advantageous because they do not cause weight gain or hypoglycemia. Despite their similar mechanism of action to GLP-1 RA, DPP-4 inhibitors do not have equivalent CV or kidney benefits.

α-Glucosidase Inhibitors

α-Glucosidase is an intestinal enzyme that digests carbohydrates by hydrolyzing complex starches to oligosaccharides in the lumen of the small intestine, releasing glucose. Inhibition of this enzyme maintains the integrity of complex carbohydrates, thereby allowing less glucose absorption. Thus, α-glucosidase inhibitors should be taken at the start of meals. These agents may be maintained but require dose adjustment with advancing CKD. Advantages of these agents include their low cost and low rates of hypoglycemia.

Insulin

Endogenously secreted insulin undergoes first-pass metabolism in the liver, leaving approximately 50% available to enter the systemic circulation. The kidney removes 30% to 80% of systemic insulin (endogenous insulin that escapes first-pass metabolism, or exogenous insulin given subcutaneously or parenterally).[35] Patients with severe CKD are prone to hypoglycemia due to reduced insulin clearance and decreased renal gluconeogenesis,[52] and doses may need to be reduced as GFR declines.[53]

Types of insulin. Commonly used rapid-acting analogs (e.g., aspart, lispro) and long-acting analogs (e.g., glargine, detemir) have a metabolic profile that is largely unaffected by CKD. Currently no specific insulin regimen is recommended in the setting of CKD. For T1D, the basal bolus regimen of 3 daily injections of short-acting insulin with meals combined with 1 or 2 injections of long-acting insulin, as used in the Diabetes Control and Complications Trial (DCCT), is the standard treatment regimen. Pumps administering continuous subcutaneous infusions of short-acting insulins are also common, increasingly paired with real-time glucose monitoring. For T2D patients requiring insulin, the regimen usually starts on once- or twice-daily long- or intermediate-acting insulin. Mixed formulations (fixed percentages of short- to long-acting insulins) or the basal bolus regimen as in T1D may be necessary if glycemic control is not achieved.

MANAGEMENT OF HYPERTENSION

Blood Pressure Goals

Blood pressure (BP) targets in patients with diabetes have gradually declined over the years. The 2021 KDIGO clinical practice guideline for the management of BP in CKD recommends the lowest target of a systolic BP less than 120 mm Hg in patients with hypertension and CKD (GRADE 2B).[54] This recommendation differs from CKD targets in other guidelines, including the 2017 ACC/AHA recommendation of less than 130/80 mm Hg, the ESC/ESH recommendation of less than 130 to 139 mm Hg systolic, the NICE target of 120 to 139 mm Hg systolic, the ADA target of less than 130/80 mm Hg (for patients with diabetes at high CV risk), and the Hypertension Canada target of less than 130/80 mm Hg.[55-58]

The lower BP target in CKD currently advocated for by the 2021 KDIGO BP guideline is largely based on a lack of heterogeneity in the CV and all-cause mortality outcomes of the Systolic Blood Pressure Intervention Trial (SPRINT) within the CKD subgroup.[59] However, the SPRINT trial, which compared less than 120 mm Hg to less than 140 mm Hg, excluded participants with diabetes, and KDIGO does acknowledge that benefits of intensive BP lowering is less certain in patients with DKD. The Action to Control Cardiovascular Risk in Diabetes (ACCORD) trial remains the preeminent trial comparing the less than 120 mm Hg to less than 140 mm Hg targets in an entirely diabetic population. This trial failed to show a difference in the primary CV outcome between groups, though the intensive target did significantly reduce the risk of stroke.[60] KDIGO argues that ACCORD excluded participants with a creatinine greater than 1.5 mg/dL (>132 μmol/L) and does not offer adequate evidence against intensive BP reduction in this population. This is contrasted with the large number of participants with CKD in SPRINT, as well as the 42% of SPRINT participants with prediabetes and preserved CV/mortality benefits within this group.[61] When considering a systolic target of less than 120 mm Hg for people with diabetes, it is important to acknowledge the lack of direct evidence supporting or refuting this recommendation.

The current recommendations of intensive BP control, whether a systolic BP less than 120 mm Hg or less than 130 mm Hg, are based on CV and mortality benefits as opposed to kidney protection. Intensive BP lowering in SPRINT, ACCORD, and the Secondary Prevention of Small Subcortical Strokes Trial (SPS3) was actually associated with higher overall rates of eGFR decline, and a meta-analysis of BP control trials including the African American Study of Kidney Disease and Hypertension (AASK) and Modification of Diet in Renal Disease (MDRD) demonstrated that more intensive BP control reduced kidney failure risk only in participants with baseline proteinuria.[62,63] Newer studies of intensive BP lowering on CV and kidney outcomes in patients with DKD are warranted, particularly in light of newer therapies offering cardiorenal protection including SGLT2i, GLP-1 receptor agonists, and nonsteroidal mineralocorticoid receptor antagonists (MRA).

Notwithstanding the recommendations of clinical practice guidelines, an individualized approach accounting for patients' unique clinical situation and preferences is explicitly recommended by KDIGO and ADA.[54,64,65] Standardized BP measurement techniques and validated equipment for all methods is an additional recommendation. Established techniques include ambulatory BP monitoring, home BP monitoring, and automated office BP measurements (preferable over auscultation).

Choice of Agents Including Renin-Angiotensin-Aldosterone System Blockade

Most clinical practice guidelines recommend the use of RAS inhibition with ACE inhibitors/ARBs in patients with DKD and moderately severely increased albuminuria.[8,58] Evidence for kidney protection from RAS inhibitors in macroalbuminuria is derived from the Irbesartan Diabetic Nephropathy Trial (IDNT) and Reduction of Endpoints in Noninsulin-Dependent Diabetes Mellitus With the Angiotensin II Antagonist Losartan (RENAAL) trials.[66,67] The use of ACE inhibitors in DKD with microalbuminuria was studied in the Micro-HOPE study demonstrating benefits in the composite CV outcome though without reductions in hard kidney outcomes.[68] The kidney benefits of RAS inhibitors in DKD patients without albuminuria are less clear. The HOPE, EUROPA, and PEACE studies do support the use of RAS inhibitors for CV protection in this population, however,[68-70] but other antihypertensive agents may confer similar cardioprotective effects in this population. The BENEDICT and ROADMAP trials additionally demonstrated a RAS inhibitor–associated reduction in the progression to microalbuminuria in patients with T2D without DKD.[71,72]

RAS inhibitor side effects include hyperkalemia, hemodynamic-mediated acute decline in kidney function, and ACE inhibitor–induced angioedema. Dual RAS inhibition with ACE inhibitors, ARB, or direct renin inhibitors is associated with increased harm from hyperkalemia and AKI without long-term CV or kidney benefits. Hyperkalemia, even if modest, may lead to premature and unnecessary discontinuation of RAS inhibitors. However, there are several approaches to mitigate these risks. In addition to dietary advice and use of diuretics, newer potassium binders such as patiromer and sodium zirconium cyclosilicate (SZC) safely reduce serum potassium levels. An ongoing trial is testing whether these agents confer CV and kidney benefits by controlling hyperkalemia to enable maximal RAS blockade in the treatment of heart failure and/or CKD (NCT03888066).

Patients with DKD usually will require more than one antihypertensive agent to achieve BP targets. Thiazide diuretics and dihydropyridine calcium channel blockers have been demonstrated to reduce CV events in large outcomes studies, which, however, were not dedicated CKD trials.[73-75] In patients with diabetes requiring antihypertensive therapy in addition to RAS inhibition, Hypertension Canada suggests that a dihydropyridine calcium channel blocker is preferable to a thiazide/thiazide-like diuretic,[58] based on favorable CV outcomes of the Avoiding Cardiovascular Events Through Combination Therapy in Patients Living With Systolic Hypertension (ACCOMPLISH) trial with benazepril/amlodipine compared with benazepril/hydrochlorothiazide.[74] The ACCOMPLISH trial was not a dedicated CKD trial and used a shorter-acting thiazide diuretic, which may have been inferior to longer-acting thiazides like indapamide or chlorthalidone. Indeed, the Action in Diabetes and Vascular Disease: Preterax and Diamicron Modified Release Controlled Evaluation (ADVANCE) trial compared combination therapy with perindopril and indapamide to usual care without thiazide-type diuretic in patients with T2D. The primary outcome—a combination of macrovascular and microvascular events that included progression of DKD—was significantly reduced in the active treatment arm. A significant reduction in "all kidney events" with active treatment was also observed. Considering that both treatment groups had high rates of RAS blockade, the observed benefits in CV and kidney outcomes were attributed to greater BP lowering and/or addition of a long-acting thiazide diuretic in the active treatment arm.[75]

ADDITIONAL APPROACHES TO SLOWING CKD PROGRESSION IN DIABETES

The significant residual risk of DKD progression despite the aforementioned therapeutic strategies has led to the emergence of additional pharmacotherapies.

SGLT2 Inhibitors

The CV and kidney protective effects of SGLT2i in patients with T2D were demonstrated in several CV outcome trials with post hoc analyses of kidney outcomes.[76-78] Two dedicated kidney outcome trials have followed: CREDENCE, involving participants with T2D, eGFR of 30 to 90 mL/min/1.73 m^2, and ACR of 300 to 5000 mg/g; and DAPA-CKD, involving participants with and without diabetes, eGFR of 25 to 75 mL/min/1.73 m^2, and ACR of 200 to 5000 mg/g. Both studies demonstrated significant reductions in kidney outcomes with SGLT2 inhibition including ESKD, decline in eGFR, and death from kidney/CV causes.[31,42] Results of DAPA-CKD were consistent among subgroups of participants with and without T2D, and SGLT2i have been incorporated into most diabetes and CKD guidelines including KDIGO.

Glucagon-Like Peptide-1 Receptor Agonists

Glucagon-like peptide-1 receptor agonists also exert significant CV benefits in T2D, though kidney benefits have been variable.[79] A meta-analysis of CV outcome trials of available GLP-1 receptor agonists did not demonstrate a reduction in the risk of doubling in serum creatinine, though there was moderate heterogeneity in the included trials.[79] In the REWIND trial comparing dulaglutide to placebo, there was an 11% reduction in the incidence of a sustained decline in eGFR of 30% or greater (hazard ratio [HR], 0.89; 95% confidence interval [CI], 0.78–1.01), but further sensitivity analyses showed a 30% and 44% reduction in the incidence of a sustained decline of eGFR of 40% or greater (HR, 0.70; 95% CI, 0.57–0.85) and 50% or greater (HR, 0.56; 95% CI, 0.41–0.76), respectively.[80] The ongoing FLOW trial (NCT03819153) involving semaglutide in participants with DKD assesses the composite outcome of persistent eGFR decline of 50% or greater, progression to ESKD, or death from kidney/CVD.

Mineralocorticoid Receptor Antagonists

Mineralocorticoid receptors are widely expressed in kidney and cardiac tissue, and their activation by aldosterone leads to hemodynamic and nonhemodynamic effects culminating in inflammation, fibrosis, and progression of cardiac and kidney disease.[81] However, clinical studies of MRA in DKD had not studied "hard" kidney outcomes and suffered from hyperkalemia events.[82,83] FIDELIO-DKD was a phase III trial of the nonsteroidal MRA finerenone, in addition to standard of care, in participants with T2D, most in CKD stage III and with a median urinary albumin-to-creatinine ratio (UACR) of 852 mg/g. Finerenone was safe and reduced rates of the primary composite of kidney failure, sustained 40% decline or greater in eGFR, or death from renal causes by 18% as well as the secondary composite CV outcome.[84] The FIGARO-DKD trial demonstrated a 13% reduction in the same composite CV outcome with finerenone compared with placebo, establishing finerenone as another treatment option to improve clinical outcomes in DKD.[84a]

Implementation of Newer Agents to Slow DKD Progression

It is expected that SGLT2i, GLP-1 receptor agonists, and nonsteroidal MRAs increasingly will be used together in the setting of DKD on a background of single-agent RAS blockade. The 28-week, phase III, DURATION-8 trial demonstrated that dual therapy with dapagliflozin and exenatide had additive effects on lowering weight and blood pressure, though with a less than additive effect on HbA1c. Additive effects on weight loss were similarly demonstrated in the SUSTAIN-9 and AWARD-10 trials.[85,86] The effect of combination therapy on heart and kidney outcomes in DKD is unknown. A propensity-matched cohort from the EXSCEL trial demonstrated that the addition of exenatide in patients on a background of SGLT2 inhibition resulted in modest improvements in major adverse cardiovascular events (MACE), all-cause mortality, and slower rates of kidney function decline compared with patients not on an SGLT2i.[87] Only 4% to 11% of FIDELIO-DKD participants were on an SGLT2i over the course of the study, and CREDENCE did not enroll participants taking an MRA at baseline, so there is limited evidence for the combined use of these two agents.

Emerging Therapies in DKD

New agents continue to be developed to prevent patients with diabetes developing CKD and progressing to ESKD (see Table 33.2). Alongside these treatment avenues is the development of more sensitive biomarkers that may aid the decision of when to commence new therapies.

Cardiovascular Complications

Diabetes is associated with increased risk for CV complications, including coronary artery disease (CAD) and heart failure. This risk is further compounded by the presence of CKD.[88-90]

Coronary Artery Disease

Although the risk of ischemic CV events is high in people with DKD, there is no established consensus about how best to detect subclinical CAD in this population. Chronic kidney disease increases CAD risk through mechanisms distinct from increased atherosclerotic disease burden. Additionally, people with CKD and CAD may present with atypical anginal symptoms. For these reasons, the usefulness of traditional CAD risk-prediction tools in people with CKD is limited.

Patients with severe CKD and stable CAD experience no benefit with coronary revascularization compared with medical therapy based on the Management of Coronary Disease in Patients with Advanced Kidney Disease (ISCHEMIA-CKD) trial, which included 57% of patients with diabetes at baseline[91] (see Chapter 85). In cases where revascularization may be indicated, patients with diabetes and severe CKD should undergo intervention irrespective of the radiocontrast agent risk in view of improved cardiac outcomes. Coronary artery bypass grafting (CABG) is superior to percutaneous coronary intervention (PCI) in patients with multivessel or complex CAD. Additionally, observational studies suggest that CABG reduces the risk from cardiac death more than PCI in DKD, including in patients on dialysis.

Treatment with SGLT2i and/or GLP1-RA should be considered in patients with T2D, CKD, and increased CVD risk, as both of these agents have been demonstrated to reduce rates of major adverse cardiac events (a composite endpoint including myocardial infarction, stroke, and CV death). In patients with severe atherosclerotic CVD, GLP1-RA may be particularly useful.

Heart Failure

People with diabetes and CKD are at increased risk for developing heart failure and related mortality and hospitalization. Factors contributing to this increased risk include higher rates of CAD, hypertension, and impaired natriuresis.

In people with heart failure with reduced ejection fraction (HFrEF), treatment with β-blockers, RAS inhibitors, MRAs, and neprilysin inhibitors reduce CV risk (see Chapter 85). Though these agents do not have proven benefits for the treatment of heart failure with preserved ejection fraction (HFpEF), they are commonly used in this population as well.

SGLT2i should be considered in subjects with increased CV risk or established HFrEF, as these agents reduce the risk of both heart failure hospitalization and heart failure progression (Table 33.2).[31,38-40,43,44] Notably, these benefits were observed even in subjects without diabetes, suggesting glycemic control is not the primary underlying mechanism. GLP1-RA use has also been associated with reduced heart failure hospitalizations in a meta-analysis of seven RCTs, though to a much lesser degree compared with SGLT2i.

Antiplatelet Agents

Hyperglycemia has a procoagulant effect on platelet aggregation independent of insulin levels, and hyperinsulinemia itself carries an inhibitory effect on fibrinolysis and thus significantly increases the risk for thrombosis. Daily aspirin therapy is recommended for secondary CV prevention in people with diabetes and a history of atherosclerotic disease. Randomized controlled trials investigating the use of aspirin for primary CV prevention have yielded mixed results, demonstrating little to no benefit on CV outcomes yet an increased risk of bleeding.[92-94] Furthermore, bleeding risk increases with advanced CKD due to uremic platelet dysfunction (see Chapter 87). As such, aspirin should be considered for primary prevention in people with diabetes and CKD only when risk of atherosclerotic CV events is high, and with careful consideration of bleeding risk. Use of dual antiplatelet agents is appropriate following acute coronary syndromes or revascularization procedures.

Dyslipidemia

Given the elevated risk for atherosclerotic events in DKD, it is generally recommended that hyperlipidemia be treated (Chapter 85). The KDIGO recommends that all adults aged 50 years or older with severe CKD receive treatment with a statin with or without ezetimibe, regardless of diabetes status. For adults with CKD who are between 18 to 49 years of age, statin treatment is recommended in the presence of diabetes, known coronary disease, or elevated CV risk. Patients should generally not be started on statins following dialysis initiation, but these agents may be continued in patients who had already been taking them prior to starting dialysis.

The effect of lowering low-density lipoprotein cholesterol (LDL-C) was studied in the Study of Heart and Renal Protection (SHARP), which included patients with nondialysis CKD and no history of CVD, randomized to simvastatin and ezetimibe or placebo. A 17% reduction in adverse CV outcomes was seen for every 33 mg/dL reduction in LDL-C regardless of diabetes status, with no impact on mortality.[95] However, in dialysis patients with diabetes, the SHARP, Rosuvastatin and Cardiovascular Events in Patients Undergoing Hemodialysis (AURORA), and Atorvastatin in Patients With Type 2 Diabetes Mellitus Undergoing Hemodialysis (4D) trials did not show significant reductions in CV events or mortality despite LDL-C falling by up to 42%.[96–98] Guidelines of KDIGO recommend continuation or cessation should be determined by the patients' condition and preference (see Chapter 85). Fibrates can reduce the risk for increased albuminuria and can replace statins for people with diabetes and eGFR less than 45 mL/min/1.73 m^2 patients who are intolerant of statins.[35] Dose reductions in statins are not normally required for advancing CKD.

Recently, proprotein convertase subtilisin/kexin type 9 (PSCK9) inhibitors, when added to maximally tolerated statin therapy, have demonstrated significant reduction in LDL-C and improvement in CV outcomes in people with and without diabetes in the Evolocumab and Clinical Outcomes in Patients with Cardiovascular Disease (FOURIER) and Alirocumab and Cardiovascular Outcomes After Acute Coronary Syndrome (ODYSSEY OUTCOMES) trials.[99,100]

MICROVASCULAR COMPLICATIONS OF DIABETES

Diabetic Foot Disease

Patients with diabetes and advanced CKD are at increased risk for diabetic foot ulcers and their subsequent complications, including infection and amputation. Foot ulcers typically occur in the setting of peripheral neuropathy, which increases susceptibility to foot injuries, and peripheral artery disease, which impairs wound healing. Patients should undergo foot examinations at least yearly consisting of visual inspection for ulcerations or other deformities, and assessment of pedal pulses and peripheral sensation. Additionally, patients should be regularly screened for symptoms of peripheral artery disease (e.g., leg fatigue, claudication). Concerns of peripheral artery disease should prompt further evaluation, such as with ankle-brachial index testing or vascular imaging. Education on appropriate footwear and monitoring for signs of ulceration and early infection are fundamental to prevention.

Retinopathy

Initial testing for diabetic retinopathy should take place 5 years after T1D diagnosis and at the time of T2D diagnosis. Subsequent testing should be performed every 1 to 2 years, or more frequently if retinopathy is present.

Maintenance of adequate glycemic control reduces the risk of diabetic retinopathy and slows retinopathy progression.[101] Therefore, glycemic control is an important therapeutic target even for patients with ESKD, for whom slowing CKD progression is no longer feasible. Ophthalmologic therapies for diabetic retinopathy include intravitreous antivascular endothelial growth factor injections and panretinal laser photocoagulation.

Neuropathy

Peripheral Neuropathy

Patients with diabetes should be evaluated for peripheral neuropathy at least annually. Patients should be assessed for symptoms of peripheral neuropathy (pain, numbness, or tingling of the extremities) and for signs of decreased temperature/pinprick and vibration sensation. The diagnosis of diabetic neuropathy should only be made after other potential etiologies of neuropathy are excluded.

Gabapentin is commonly used to treat neuropathic pain associated with diabetes, but dosing should be reduced as GFR declines. Dosing may require alternate days or administration after hemodialysis sessions to avoid the known sedative and motor (myoclonus) effects of gabapentin accumulation that occur in individuals with reduced eGFR. Pregabalin is an alternative that is predominantly excreted by the kidneys and should be used cautiously with advancing CKD.

Gastroparesis

Gastroparesis occurs commonly in people with diabetes, can affect the absorption of oral medication, and can contribute to volume depletion due to poor oral intake. Pharmaceutical options for management of gastroparesis include metoclopramide, domperidone, and erythromycin. A surgically implanted gastric pacemaker may also alleviate symptoms in patients with severe symptoms. Optimizing the management of gastroparesis may be very challenging but should be considered before transplantation in view of the potential effects on immunosuppressant absorption.

Autonomic Neuropathy

Diabetic autonomic neuropathy may result in significant postural hypotension and orthostatic symptoms. As such, patients may require changes to their antihypertensive regimens in order to reduce hypotension risk. Additionally, autonomic neuropathy may contribute to hypotension during hemodialysis, necessitating the use of agents such as midodrine. Midodrine is contraindicated in patients with arrhythmias and previous cardiac events and, if used, should be done so with caution. Nonpharmaceutical approaches to managing postural hypotension include liberalizing dietary sodium and utilizing compressive leg or abdominal garments.

Erectile Dysfunction

Erectile dysfunction frequently occurs in DKD as a combination of vasculopathy and side effects of medications used to treat CAD and hypertension. Phosphodiesterase inhibitors may be used unless the patient is taking nitrates or has uncontrolled CAD.

COMPLICATIONS OF CHRONIC KIDNEY DISEASE

Anemia

Anemia occurs commonly in DKD and acts as a risk multiplier for all-cause mortality and an independent risk factor for left ventricular hypertrophy (LVH), CVD, and congestive cardiac failure (Chapter 86). Compared with nondiabetics, diabetic patients have lower hemoglobin (Hb) levels at every CKD stage. One study found the prevalence of anemia to be 41% in diabetic versus 17% in nondiabetic CKD patients, occurring before kidney function begins to decline.[102] Therefore, patients with diabetes and eGFR less than 60 mL/min/1.73 m^2 should be tested for anemia.[103] Patients with DKD have increased levels of proinflammatory cytokines from low-grade systemic inflammation, causing resistance to the effects of erythropoietin in various tissues of the body that impairs the efficient use of iron in the generation of new erythrocytes. Other factors causing resistance include autonomic neuropathy, RASi, and bone marrow microvascular injury.

The Trial to Reduce Cardiovascular Events With Aranesp Treatment (TREAT) assessed the use of erythropoiesis-stimulating agents in patients with T2D with GFR 20 to 60 mL/min/1.73 m^2 and Hb 11 g/dL or less. Patients were randomized either to darbepoetin-alfa (targeting a Hb of 13 g/dL) or control with rescue darbepoetin treatment if Hb was less than 9 g/dL. The treatment group had fewer CV revascularization procedures but significantly increased risk for fatal and nonfatal nonhemorrhagic strokes.[104] Of the TREAT cohort, 31% progressed to dialysis and death despite good BP, glycemic, and lipid control.

Long-term CV and kidney benefits of prolyl hydroxylase inhibitors (PHIs), which effectively increase Hb in DKD, have yet to be determined (see also Chapter 86).[105] In an exploratory and pooled analysis, roxadustat led to fewer CV outcomes (a composite of myocardial infarction, stroke, or all-cause mortality) than epoetin alfa in patients new to dialysis (with and without diabetes) (HR, 0.70; 95% CI, 0.51–0.96).[106] The rationale for the use of PHIs in DKD includes kidney hypoxia induced by DKD leading to HIF-1α activation. However, DKD may also be associated with instability of HIF-1 α with loss of downstream compensatory pathways to hypoxia and subsequent upregulation of pathogenic pathways.[107] Prolyl hydroxylase inhibitors have been demonstrated in animal models to ameliorate deleterious metabolic and transcriptional changes associated with hypoxia in DKD.[107]

Mineral Bone Disease

Imbalance in mineral metabolism increases CV risk and is associated with bone disease in DKD. Targets for parathyroid hormone (PTH), calcium and phosphate levels, along with management should be aligned to those with non-DKD (Chapter 88).

Electrolytes and Fluid Retention

Salt restriction and diuretics are the mainstay treatment of fluid retention in DKD. Metabolic acidosis can be treated with bicarbonate supplementation. In patients with hyperkalemia, diet and RASi should be reviewed, and treatment with diuretics, sodium bicarbonate, gastrointestinal potassium binders, or dose reduction or discontinuation of RASi should be considered.

END-STAGE KIDNEY DISEASE

Decisions regarding dialysis modality should be guided by patient preference. There is no evidence to suggest superiority of one dialysis modality over another in people with diabetes and kidney failure. Mortality risk in this population is high and concomitant with the presence of multiple comorbidities.

Dialysis

Hemodialysis

Considerations for vascular access placement, hemodialysis initiation, and hemodialysis prescription in DKD are similar to those in patients without diabetes.

Peritoneal Dialysis

Most peritoneal dialysis solutions utilize dextrose as an osmotic agent. The dextrose content of peritoneal dialysis solutions may worsen hyperglycemia in people with diabetes. As a result, patients may need to adapt their hypoglycemic drug regimens. Commonly used peritoneal dialysate solutions contain between 1.5 to 4.25 g/dL glucose and may provide an additional 400 to 800 kcal/day.

Non–glucose-containing solutions (e.g., icodextrin) avoid the risk for hyperglycemia and the potentially harmful effects of glucose degradation products while resulting in better ultrafiltration volume, BP control, and preservation of residual function (Chapter 102). Some metabolites of icodextrin also can act as a substrate for the glucose dehydrogenase present in blood glucose meters, resulting in potential overestimation of blood glucose levels.

Transplantation

Diabetic kidney disease remains the leading cause of ESKD among waitlisted patients. Kidney transplantation confers mortality benefit in these patients: the US Renal Data System data showed 5-year survival rates of 29% in patients with diabetes starting dialysis compared with rates of 75% and 85% in those undergoing deceased-donor and living-donor kidney transplants, respectively.[108] When possible, preemptive kidney transplantation and living donor transplantation would be preferable to transplantation after starting dialysis and deceased donor transplantation, respectively. Considering the significant risk of remaining on dialysis and the benefit conferred by even marginal deceased donor kidney allograft, many patients with diabetes and ESKD should consider consenting for kidneys with a high kidney donor profile index (≥85%). For eligible patients with T1D, the ideal form of transplantation is simultaneous pancreas and kidney transplantation, or living-donor kidney transplant followed by deceased-donor pancreas transplantation.[35]

SELF-ASSESSMENT QUESTIONS

1. A 56-year-old man with T2D, eGFR of 38 mL/min/1.73 m^2, peripheral artery disease, and a history of myocardial infarction is referred for worsening proteinuria. His BP is 133/82 mm Hg, and his body mass index is 32 kg/m^2. His HbA1c is 9.3% using canagliflozin and metformin. Which of the following changes to his antihyperglycemic regimen should you recommend?
   A. Replace canagliflozin with empagliflozin.
   B. Replace metformin with extended release exenatide.
   C. Add liraglutide to current regimen.
   D. Add linagliptin to current regimen.
   E. Add pioglitazone to current regimen.
2. Which of the following factors favors a higher (more conservative) individualized HbA1c target?
   A. History of recurrent severe hypoglycemia
   B. Availability of continuous glucose monitoring
   C. Absence of other diabetes microvascular complications
   D. Higher eGFR
   E. Relatively few comorbidities
3. Which of the following statements regarding BP control in diabetes is *false?*
   A. There is evidence of a reduction in the risk of stroke with an intensive systolic BP target of less than 120 mm Hg versus less than 140 mm Hg in patients with diabetes.
   B. Intensive BP lowering in the SPRINT trial was associated with lower rates of eGFR decline.
   C. Automated office BP measurements are preferable to those made using the auscultation technique.
   D. The 2021 KDIGO clinical practice guideline for the management of BP in CKD makes a grade 2B recommendation to target a systolic BP of less than 120 mm Hg in patients with hypertension and CKD.
   E. The strongest evidence for kidney protection with RAS inhibition exists in patients with DKD and severely increased albuminuria.
4. Which of the following is *true* regarding newer therapies with cardiorenal protection in DKD?
   A. The DAPA-CKD trial exclusively enrolled participants with T2D, eGFR of 30 to 90 mL/min/1.73 m^2, and an albumin-to-creatinine ratio of 300 to 5000 mg/g, demonstrating the kidney benefits of dapagliflozin.
   B. The CREDENCE trial demonstrated the kidney benefits of canagliflozin in participants with and without diabetes, eGFR of 25 to 75 mL/min/1.73 m^2, and albumin-to-creatinine ratio of 200 to 5000 mg/g.
   C. In a subgroup analysis within DAPA-CKD, the effects of dapagliflozin with respect to kidney protection were greater in participants with diabetes compared with those without.
   D. The cardiovascular benefits of SGLT2i are largely related to reductions in heart failure hospitalizations, whereas there is stronger evidence of reductions in atherosclerotic cardiovascular disease with GLP-1 receptor agonists.
   E. The kidney benefits of finerenone in the FIDELIO-DKD trial were demonstrated in a population with high rates of RAS and SGLT2 inhibition.

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